

TAARALA THOTA

A Garden Of Stars



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Taarala Thota · Garden of Stars

A calm, ad-free garden of games and a private shortcut to the people who matter, built as an installable web app. Live at taarala-thota.pages.dev · October 2026

This document is written for product, design and engineering readers. Each section stands on its own; use the outline to jump between them.

About This Project

Taarala Thota is a personal, non-commercial learning project. It's free, has no ads and isn't for sale. I've worked hard to protect privacy and get things right, but I'm not a legal or security professional, and there may be mistakes. If you notice anything that could cause harm, please tell me through Share Feedback and I'll fix it as quickly as I can.

Index

1	Product Case Study
2	Product Requirements Document (PRD)
3	AI-Assisted Development Methodology
4	Product Roadmap and Prioritization
5	UX Design and User Journeys
6	Product Architecture and Decision Records
7	QA and Validation Report
8	Privacy, Security and Risk Assessment
9	Product Decision Log

1. Product Case Study

At a Glance

- **Try it:** taarala-thota.pages.dev (best on a phone)
- **See it:** a 1½-minute tour video, in the README
- **Code and history:** github.com/rdg-git/taarala-thota

Key decisions

1. **Everything stays on the phone:** no accounts and no server for personal data; locked backups instead of cloud sync.
2. **Settings protected by a passcode:** games and loved ones stay one tap away, numbers stay private.
3. **The person chooses when to update,** after seeing exactly what's new; never during play.
4. **Anonymous visit counting with a clear notice,** an easy off switch, and Global Privacy Control respected.
5. **Nothing reaches the live app untested:** test branch, protected preview, and my approval of every change and its wording.
6. **A playing tour on the real screens** instead of a manual: a few KB, works offline, never out of date.

Summary

Product	Taarala Thota (Garden of Stars): An installable web app (Progressive Web Application) with four classic games with gentle beginner levels, a private "Loved Ones" shortcut to message or call chosen people on WhatsApp, and a built-in guided tour (Take A Tour).
Platform	Installable web app (PWA): works in any phone browser, installs to the Home Screen, works offline
My role	Product manager: ideation, problem, requirements, every product, privacy and design decision, acceptance and release approval
How it was built	AI-assisted: Claude (Anthropic) wrote the code to my requirements; ChatGPT reviewed the work as a second model (inspired by maker-checker); I decided
Timeline	Private family version from February 2026; public version complete 23 Oct 2026; live 28 Oct 2026; eleven approved releases by 29 Oct 2026
Quality	29 automated test suites (more than 420 checks), a security review with an attack test, and real-phone testing on a protected preview before release

Background

Taarala Thota started as a public offshoot of a private family app I'd been building since February 2026. I was looking for safe, calm games for my family and kept finding free apps full of ads and banners, where one accidental tap opened a pop-up, a video or a store page, which often led to another. I wanted a space where that simply couldn't happen. The idea then grew to include a focused way to reach a few chosen loved ones. Taarala Thota is that idea, made public: no ads, no sign-in, nothing personal collected.

The problem

Free game apps are rarely calm. Most are paid for by advertising, and a single accidental tap opens a pop-up, a video or an app-store page. Many ask for an account and collect personal data. At the same time, reaching the people who matter most usually means digging through a crowded contact list. This was my family's need.

What I knew and what I assumed

- **Observed first-hand:** before building, I downloaded several free games from the Google Play Store, mostly bubble-pop games and piano games that turn the screen into keys. Every one had either sudden pop-ups that could be tapped by accident or ad banners, so I could never hand the phone over and walk away.
- **Assumed:** other people want the same calm, ad-free space.
- **Assumed:** a short, private list of loved ones is valuable alongside the games.
- **Not yet validated:** whether people outside my family find it useful, and which games matter most to them.
- **How I'll validate:** feedback after launch, and watching a few people try it for the first time (see Success measures).

This is a personal, non-commercial project, so I didn't run formal customer research. I've kept what I observed separate from what I assumed.

The solution

A garden that opens straight onto two doors, **Play** and **Family**:

- **Play**: Bubble Pop, Hidden Comets (a gentle take on minesweeper), Chess and International Checkers. Every game has easy beginner levels; Chess and Checkers offer hints and undo on Easy, so anyone can learn.
- **Family (Loved Ones)**: up to 20 chosen people as large circles with photos. One tap opens WhatsApp to message or call.
- **Privacy:** Nothing to sign up for, nothing personal collected: no account, no ads, no personal tracking. Everything you save stays on the phone, protected by a Settings passcode and optional password-locked backups. Visits are counted anonymously (no cookies) and can be switched off.
- **Steady**: stray taps, long presses and swipes never throw you out of the app.

- **Take A Tour**: a 1½-minute walkthrough that plays on the real screens and shows where to tap, with pause, speed and a bar to go back or forward.



Choose Your Adventure · Chess on Easy with move hints · Privacy & Safety

Outcome

- Live and installable, with a test-then-release process: every change is tested on a protected preview before it reaches the live app, and people see exactly "What's New" before they choose to update.
- A product with no personal data to leak by design apart from anything people choose to type into feedback (see Feedback content), an honest privacy page, and documented, disclosed limitations.
- A public repository showing the full history of decisions, tests and releases.
- Anonymous visit counts and an in-app feedback box show whether it's working, without tracking anyone.

What I learned

AI made execution dramatically faster, but quality came from the product discipline around it: clear requirements, explicit trade-offs, strict acceptance criteria, an independent checker, and testing on real devices. Two of the most important defects were found by my own real-world use after all automated tests had passed.

2. Product Requirements Document (PRD)

Goals

1. Offer calm, constructive play with no ads, no sign-up and no personal tracking.
2. Put the people who matter one tap away, privately.
3. Keep every piece of personal data on the person's own phone, under their control.
4. Be usable by anyone, including beginners, without instructions.

Non-goals (for this version)

- Accounts, cloud sync or server-side storage of personal data
- Playing against other people online.
- Advertising of any kind.
- App-store distribution (it is an installable web app).
- Personal tracking or analytics of what people do inside the app.

Who it is for

Person	What they need	Focus
Someone who wants a few good games without ads	Open and play in two taps; nothing interrupts	Primary
A beginner learning chess or checkers	Easy levels, move hints, undo	Secondary
Someone who wants their closest people one tap away	A short, private list that opens WhatsApp directly	Primary
Someone sharing their phone with another person	Settings and numbers protected; the screen stays in the app	Secondary
Someone new to the app	Take A Tour shows where to tap in about 1½ minutes	Secondary

The primary focus is calm play and reaching loved ones quickly, for myself and my family first. The other needs are secondary, and the app stays open to everyone.

Functional requirements and acceptance criteria

#	Requirement	Accepted when
F1	Open straight to Home, no sign-in	First visit shows Home with Play and Family; no prompts
F2	Four games with beginner levels	Each game playable end to end; Bubble Pop levels all open; accuracy goals earn gold stars
F3	Hints and undo on	Hints/undo appear on Easy in Chess and Checkers, never on

#	Requirement	Accepted when
	Easy only	harder levels
F4	Loved Ones: one tap to WhatsApp	Tapping a person opens WhatsApp for that number; nothing is sent automatically
F5	Settings protected by a passcode	Numbers can only be seen or changed after the passcode; Settings locks when you leave it and after 10 minutes without a tap in Settings
F6	Forgot passcode	Typing ERASE removes saved contacts and the passcode only; games and progress stay
F7	Backup and restore	Backup creates a password-locked file; restore checks every item and keeps the current passcode
F8	Backup reminder	Shown on Home only, every 6 days once there is something to save
F9	Works offline	After the first visit, the app opens and plays without internet
F10	Updates never interrupt	A new version waits; the person sees “What’s New” and chooses Update Now or Later; a small “Update” button and Settings let them update any time; never during play
F11	Privacy & Safety page	Plain-language points, including what the hosting service handles, how to keep data safe, and the official address
F12	Laptop visitors	A welcome card offers a QR code to open on a phone, or Continue On This Screen
F13	Phone turned sideways	A calm “turn upright” card appears and Bubble Pop pauses
F14	Anonymous visit counting	Counted only on the official address and only after the one-time welcome note; switch on Privacy & Safety and in Settings, every change confirmed; never counted when the browser sends Global Privacy Control
F15	Anonymous feedback	One box from Privacy & Safety or Settings; only the message, the app version and a one-time anti-spam token are sent; 10,000-character limit; daily cap
F16	Take A Tour	Starts from the top of Privacy & Safety and from Settings; plays on the real screens with pause, 5-second back/forward, drag bar, speed and music; nothing is tapped or saved; install steps match the phone

Non-functional requirements

Area	Requirement
Privacy	Nothing you save leaves the phone; no cookies; visit counts are anonymous and can be switched off; feedback is anonymous
Security	Strict browser security rules; the only outside code is Cloudflare's

Area	Requirement
	anonymous counting script (loaded only when counting is on) and Cloudflare's Turnstile anti-spam check (loaded only on the feedback box); passcode never stored, only a fingerprint; backups encrypted
Reliability	Data kept across updates; app reopens reliably online and offline
Performance	About 1 MB to open Home; pictures in modern compressed formats
Accessibility	Finger-sized buttons (44 px minimum), screen-reader labels, reduced motion respected, two-finger zoom on text pages
Maintainability	Plain HTML/CSS/JavaScript, inline code notes, a developer guide, automated tests

Success measures

Success is measured with the lightest possible footprint: anonymous counts and voluntary feedback, never personal tracking.

- **Reach (anonymous):** visits, where they came from (for example LinkedIn or GitHub), countries and device types, via Cloudflare Web Analytics. No cookies, no unique-user tracking, one count per app opening, never in-app behaviour. People can switch counting off, and browsers sending Global Privacy Control are never counted.
- **Voice of the user:** anonymous feedback from the in-app “Share Feedback” box: what people enjoy, what confuses them, and what they’d like next.
- **Quality gates:** every release passes all automated tests, is checked on a protected preview, and is approved (changes and What’s New text) before going live.
- **Privacy guardrail:** nothing people save in the app is held anywhere other than their own phone. Visit counts and feedback are anonymous by design.

Deliberately not measured: unique users, time in app, which games are played, retention, or anything a person saves.

Trade-off accepted: approximate numbers (visits rather than people; offline opens and opted-out visitors aren’t counted), in exchange for a privacy promise people can keep control of.

What I want to learn:

Hypothesis	How I’ll check (without tracking anyone)	Status
A first-time visitor can open the app and start a game without help	Watch 3–5 people try it with no instructions; note where they hesitate	Not yet tested

Hypothesis	How I'll check (without tracking anyone)	Status
People can add a loved one and reach them in one tap	In the same session, ask them to add a test contact and open it	Not yet tested
People understand that their data stays on their phone	After they read Privacy & Safety, ask them to explain it in their own words	Not yet tested
No stray tap takes people out of the app	Note any accidental exits in the sessions; feedback themes	Tested in my own use and automated tests; not yet with others

Usage isn't tracked inside the app, so these are checked by watching and asking, not by analytics.

3. AI-Assisted Development Methodology

Tools and roles

Tool	Role
Gemini	Early guidance while I shaped the idea
Cursor	AI code editor used in early experiments; I dropped it once I found working directly with Claude and the Windows terminal simpler
Claude (Anthropic)	The **maker**: wrote the code, the automated tests and the technical documentation; some of the artwork
ChatGPT	Second-model review: I sent it each release package to look for problems; also used for brainstorming and to create most of the painted artwork
Me	Product owner: requirements, decisions, acceptance, release approval

I borrowed the **maker-checker** control from my experience as a Product Manager in the financial services industry: one AI built, a second AI reviewed, and I made the final decision. It is inspired by that control, not equivalent to an audited one.

Limits of this approach

- Both AI models can be wrong in similar ways, so a second model is not a guarantee.
- The automated tests were written by the same AI that wrote the code.
- The truly independent controls were my own testing on real devices and my approval of every release.
- None of this is a professional security, privacy or legal certification.

The working agreement I set

Before any code was written, I set rules that the AI followed throughout:

- Explain everything in plain language; never assume.
- For every choice, show the options, the trade-offs and the implications (security or otherwise).
- Ask before implementing anything.
- Never put personal names, emails, phone numbers or keys in anything that could become public.
- Source control one step at a time; nothing published or merged without my explicit OK.
- Update documentation only when I ask.
- Before every release, show me the exact “What’s New” text and wait for my approval.
- Brainstorm and show mockups first; change the app only after I confirm.

How a change moved from idea to release

1. **Requirement**, stated as an outcome (for example: “a stray tap must never leave the app”).
2. **Options and trade-offs** presented by the AI, often with a recommendation.
3. **My decision**, sometimes different from the recommendation.
4. **Build and automated tests** by the AI, including new tests for the change.
5. **Visual review**: screenshots on several phone sizes; I accepted, rejected or asked for changes.
6. **Release approval**: I approved the changes and the “What’s New” wording.
7. **Test on a protected preview**, then **merge** to publish (see section 6, A14).

Examples of directing and correcting the work

- **Rejected designs**: I called the first “turn your phone upright” card an eyesore and had it redesigned in the garden’s style. I asked for new rounds of maker’s-mark designs until one felt personal, then chose to combine three. I also brainstormed extensively with ChatGPT and Claude to create and refine the designs, aesthetics and each screen layout.
- **Overrode a recommendation**: the AI suggested capping “What’s New” at the last five releases; I chose no cap (“those who read will be particular”) with slightly smaller text. This is especially important in cases where people have not updated the app in a long time. This ensures transparency and what all changed.
- **Changed the requirement after real use**: after tapping “Later” on an update, I couldn’t find how to update again, so I added an update section in Settings and a small “Update” button, visible only while an update waits.
- **Caught what tests missed**: the live host’s redirect broke reopening the app, and the laptop card did not appear on my touch-screen laptop. Both passed automated tests; both were found by my own use.
- **Raised a security gap**: I noticed Settings stayed unlocked unless “Lock” was pressed. It now locks the moment you leave it.
- **Redirected a feature after seeing it**: the first guided tour was step-by-step cards. I asked for a playing walkthrough, like a screen recording, with pause, a drag bar and speed, and removed wording that pointed people towards opting out of counting.
- **Chose a notice over a question** for visit counting: A clear one-time note and an easy switch respect people without hiding anything.
- **Set a consistency rule**: after noticing “Take a tour” beside “What’s New”, I had every button and title moved to Title Case.
- **Shaped the privacy notices:** I wanted people told clearly where their data lives. My first idea, a pop-up every time, became a calmer design: a short note in Settings, a once-a-day reminder, and a plain-language Privacy & Safety page that separates what the app keeps from what any website’s host handles.

What I kept ownership of

Every product, privacy, wording and design decision; what went into each release; and the final call on every trade-off. The AI proposed; I decided.

Lessons

- AI is fastest when requirements are outcomes and constraints are explicit up front.
- Asking for options and trade-offs, not answers, kept decisions with me.
- An independent checker and real-device testing are essential: automated tests only test what someone thought of.
- Fewer tools, used well, beat more tools: I dropped an AI code editor once working directly in the terminal proved simpler.

4. Product Roadmap and Prioritization

How I prioritized

1. **Trust first**: anything touching privacy or security came before features.
2. **The core promises next**: calm play and one-tap connection.
3. **Polish from real use**: changes driven by what I hit while using it.
4. **Defer anything that needs a server for personal data**, because it changes the privacy model.

Core scope (the first complete version)

Four games, Loved Ones with WhatsApp, Settings passcode, Privacy & Safety page, steady screen, offline use, strict security rules.

Added in increments (roughly in order)

Release	What was added	Why
Backups	Password-locked backup and restore, 6-day reminder	Data lives only on the phone; people need a way to keep it
Contact photos	Optional photos, redrawn by the app	Faces are faster to recognize than names
Screen fill	Pictures fill wide phones without side strips	Looked unfinished on many phones
Hints on Easy only	Hints and undo limited to Easy	Help beginners without making harder levels trivial
Turn-upright card	Calm card when a phone turns sideways; Bubble Pop pauses	The garden is painted for upright screens
Ownership marks	Maker's seal, signature, code notices, invisible artwork watermark	Clear authorship before going public
Laptop card	"Best On Your Phone" with a QR code	Many visitors (including from a CV link) arrive on a laptop
Open levels	All Bubble Pop levels open; gold stars for goals	Match Chess and Checkers; let anyone try anything
What's New	Release notes before updating; update button and Settings section	People should know what changes before they accept it
Auto-lock	Settings locks when you leave it	Close a real gap on shared phones

Release	What was added	Why
Clearer data advice	Privacy page on keeping data safe, picture download, zoom	Honest expectations; accessibility
Official address	The one official address on the Privacy page	Protect people from copies
Visit counting	Anonymous counts, a one-time welcome note and an on/off switch	Know roughly how many people visit, without tracking anyone
Feedback	Anonymous Share Feedback box	Hear from people directly
Take A Tour	A playing walkthrough with player controls	Show new people where to tap, without a manual
Consistent capitals	Title Case on every button and title	Polish and consistency
App icon and artwork	A new painted app icon; the Hidden Comets tile now shows a comet and a star instead of a mine and a flag	A clear icon at thumb size; artwork that matches the game's theme
Fit, encouragement and review	Full-screen fit on tall phones; a "good move" cheer in Chess and Checkers; a replay to review a finished game; a hidden honeypot and Cloudflare Turnstile to block automated feedback spam	Comfort touches that make the games friendlier, and keep the feedback box clear for real people

Deliberately deferred

Item	Why deferred
Accounts / cloud sync	Requires a server and personal data; backups cover the need
Detailed usage analytics	Every metric is data about someone; only anonymous visit counts are kept
Fingerprint / face unlock	A passcode was enough for the MVP

Next (still in idea phase)

- Friendly challenges between people in Loved Ones. The direction I most want to explore, but it changes the privacy model: data would leave the phone, so it needs a server design and a fresh privacy review first.
- A “larger text” setting, building on zoom for text pages.
- More games, videos based on ancient wisdom, etc.

Launch plan

- Before going public: make the GitHub repository public with branch protection and secret scanning, delete the old test site, and run a final personal-information scan.

- GitHub README: screenshots, the tour video, how to install and use it, and a link to this document.
- First 30 days: watch visits and where they come from, read feedback, and fix anything that could cause harm first.

5. UX Design and User Journeys

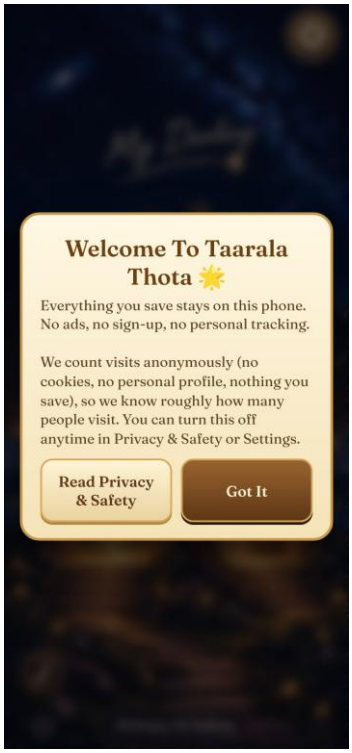
Design principles

- Nothing between you and the game: no sign-in, no splash, no permission prompts.
- Calm by default: soft music made live in the app, a painted night garden, no flashing ads.
- Stray taps do nothing harmful: no long-press menus, no accidental zoom or swipe out of the app.
- Honest and plain: every message in everyday words; limits explained rather than hidden.
- The person decides: updates wait for them; nothing happens automatically that they can't see.

Key journeys

Journey	Steps	Design decisions
First open	Address → Home	Opens straight onto Home; on a laptop, a card offers the phone instead. A one-time welcome note explains privacy and anonymous counting.
Play	Play door → Games → game → level	Every level open; Easy shows hints and undo; gold stars for goals
Reach someone	Family door → tap a circle	Opens WhatsApp; nothing sent automatically; numbers never shown here
Add a loved one	Gear → passcode → Add A Contact	Photo redrawn by the app; numbers checked (+ and country code)
Keep data safe	Settings → Back Up Now	Password-locked file saved wherever the person chooses; reminder every 6 days
Restore	Settings → Restore From A Backup	Confirmation shows date and contact count before replacing anything
Update	"Update" button or card → What's New → Update Now / Later	Only releases newer than the installed one; never during play
Turn sideways	Phone rotates	Calm card; Bubble Pop pauses; nothing lost

Journey	Steps	Design decisions
Share the phone	Privacy & Safety tip	Guided Access (iPhone) / App pinning (Android) explained
Learn the app	Privacy & Safety or Settings → Take A Tour	Plays on the real screens; pause, drag, speed; nothing is tapped or saved
Share feedback	Privacy & Safety or Settings → Share Feedback (Anonymous)	One box; no name or email
Review a game	Game ends → Review Game → step through the moves	Replays the game's moves, which stay on the device with the resumable game, on a clean board; the walkthrough is not stored, and Back or Home clears it





First open: a one-time welcome note · Take A Tour playing, with its player bar

Settings at a glance

- App update status sits first: "✓ Your app is up to date", or a compact "New Update Available" with Update Now, Update Later and What's New. Then come the Take A Tour card, the visit-counting switch and Share Feedback (Anonymous), and then the passcode screen. Settings locks itself as soon as you leave it, and after 10 minutes without a tap.

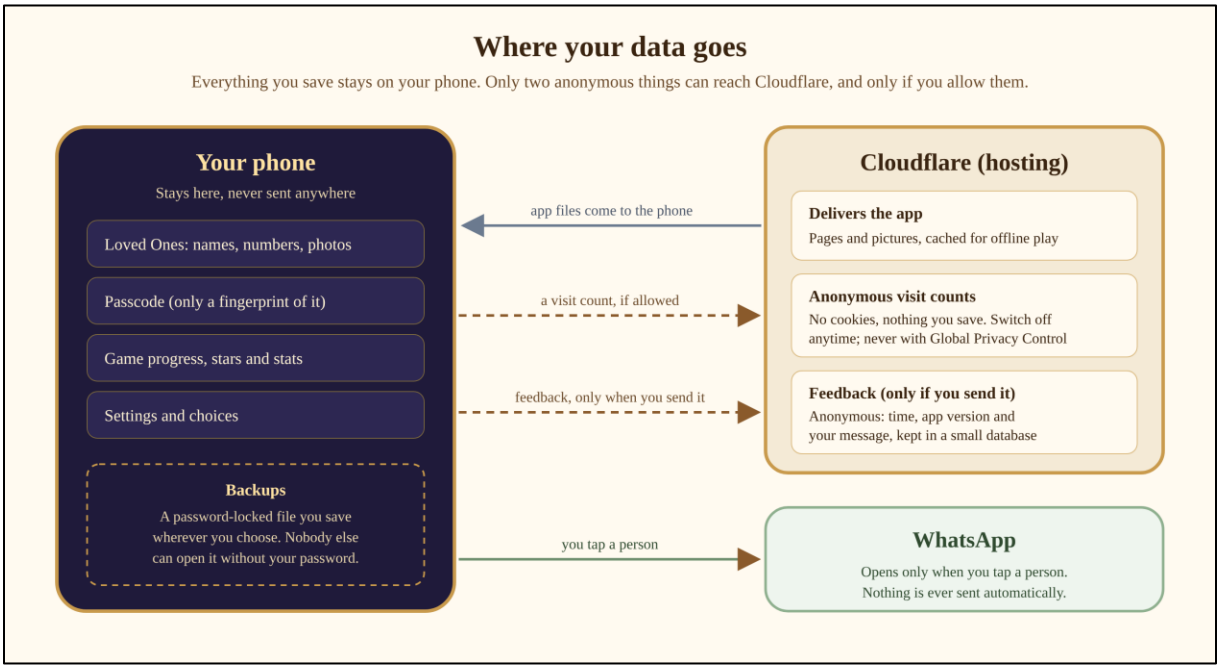
Accessibility

- Buttons at least 44 px, even on small screens.
- Screen-reader labels on buttons and icons; decorative drawings hidden from screen readers.
- Reduced-motion settings respected.
- Two-finger zoom on the text pages (Settings, Privacy & Safety); games stay steady.
- Trade-off accepted: zoom stays off on game screens to protect against stray touches.
- Take A Tour can be paused, slowed to 0.5× and dragged back to any point.

6. Product Architecture and Decision Records

Architecture in brief

A set of plain HTML, CSS and JavaScript files served by Cloudflare Pages. Each screen is a painted picture; buttons and words are placed on top in the picture's own coordinates, so layouts scale to any phone. Everything a person saves stays in their browser's storage. An offline helper (service worker) keeps the app's own files for offline use and manages updates. There is no server for personal data. One small Cloudflare function receives anonymous feedback and stores it in a Cloudflare D1 database (time, app version and message only; read from time to time; abusive messages may be deleted). The only outside code is Cloudflare's anonymous visit-counting script (loaded only on the official address and only when counting is on) and Cloudflare's Turnstile anti-spam check (loaded only on the feedback box, to block automated spam).



Where your data goes: everything you save stays on the phone

Decision records

#	Decision	Options considered	Chosen	Trade-off
A1	Technology	Framework · plain web code	Plain HTML/CSS/JavaScript, one small build step	Less tooling; fewer dependencies and nothing third-party to trust

#	Decision	Options considered	Chosen	Trade-off
A2	Where data lives	Server database · phone storage	Browser storage on the phone (labels start `tt:`)	No sync between devices; nothing to leak or breach
A3	Passcode	Store it · store a fingerprint	Salted fingerprint only (PBKDF2-SHA256, 210,000 rounds); growing wait after 5 wrong tries	A forgotten passcode can't be recovered (contacts must be re-added)
A4	Backups	Cloud · plain file · encrypted file	Encrypted file (AES-GCM 256, key from the backup password)	Lost password = unusable backup
A5	Restoring	Trust the file · check everything	Treat every file as untrusted: size, structure, allowed items, every contact re-checked	Slightly more code; hostile files are refused
A6	Offline and updates	Always online · offline helper	Offline helper; updates download quietly and apply only when the person chooses	More moving parts; tested specifically
A7	Release notes	Fetch from a server · carry inside the update	Carried inside each new version; shows only releases newer than the installed one	Notes must be written at release time (and approved)
A8	Security rules	Defaults · strict policy	Strict Content Security Policy and related headers: only the app's own files; plus, Cloudflare's counting script and its reporting address, and Cloudflare's Turnstile anti-spam check on the feedback box (the only framing allowed); no camera, mic or location	Inline styles still allowed (low risk; documented)
A9	Pictures	One format · modern with fallback	AVIF with WebP fallback, fingerprinted addresses for long caching	Two copies of each picture in the project
A10	Fonts	Google Fonts · bundled	Bundled fonts (open licence)	Slightly larger app; no outside connection
A11	QR code on laptops	Online QR service · drawn in the app	Drawn by the app	A little more code; no outside service sees the address

#	Decision	Options considered	Chosen	Trade-off
A12	Computer player	On the main screen · background helper	Background helper (web worker)	Keeps the screen smooth during the computer's turn
A13	Hosting	Several free hosts	Cloudflare Pages, connected to GitHub	The `.pages.dev` address ties the app to Cloudflare
A14	Releases	Publish on every change · test first	Test branch → protected preview → merge	A few extra steps per release; nothing untested reaches the live app
A15	Visit counting	None · Google Analytics · Cloudflare Web Analytics	Cloudflare Web Analytics, loaded by the app only when allowed	Approximate numbers; one outside script
A16	Feedback	Google Form · email link · own function	Own Cloudflare function + D1, accepting messages only from the app, with a daily cap	A small server piece to maintain
A17	Guided tour	Video file · step cards · live replay	Live replay on the real screens (a few KB, works offline, never out of date), recorded to MP4 for social media	More code than a video file

Running cost

$0 a month (hosting only; excludes AI subscriptions and my time). Everything runs on Cloudflare's free plan (checked October 2026):

- Hosting (Pages): the app's files are served free and without limit; up to 500 builds (releases) a month.
- Feedback function: shares a free allowance of 100,000 requests a day; only sending feedback uses it.
- Feedback database (D1): 5 GB of storage, 100,000 writes and 5 million reads a day; each message is one write.
- Visit counting (Web Analytics): included on the free plan.
- If a limit were ever reached, only feedback would pause until midnight UTC. The games and Loved Ones keep working, because they run on the phone. Free limits can change, so I re-check them before major releases.

Dependencies

Depends on	Used for	If it changes or fails	Fallback
Cloudflare Pages	Hosting	New visitors can't load the app; installed copies keep working offline	The same files can be hosted elsewhere (e.g. GitHub Pages); the address would change
Cloudflare Web Analytics	Visit counts	Counts stop; the app is unaffected	Switch it off or use another privacy-friendly counter
Cloudflare Functions + D1	Feedback	Sending fails with a friendly message; the text stays in the box	Pause feedback or move it to another small service
WhatsApp link format (wa.me)	Loved Ones	Taps stop opening chats	The link is built in one place in the code, so it's a quick change
GitHub	Code, history, releases	Releases pause	Upload directly to Cloudflare; a full copy is kept locally

- Fonts, pictures and the QR code are all inside the app; nothing else is loaded from outside.

7. QA and Validation Report

Test strategy

- Automated browser tests for every flow, run under the live security rules on a local server that mimics the host.
- New tests for every change and every defect, so it can't come back.
- Visual review on several phone sizes, from a small Android phone to a tablet.
- Real-device testing before release, on a protected preview of the exact release.
- Second-model review of each release package by ChatGPT.

Coverage (29 suites, more than 420 checks)

Area	What is checked
Navigation and games	Every screen and game flow; Bubble Pop levels and stars; full chess rules against standard move-count tables; checkers rules
Contacts and passcode	Create, unlock, wrong tries and waits, forgot passcode, auto-lock when leaving, 10-minute timeout
Backups	Create, restore, wrong password, tampered, fake and oversized files, hostile contents
Offline and updates	Offline play, reopening, update waiting, What's New (only newer releases), Update button visibility, Settings update box
Security and privacy	App works under strict rules; no unexpected outside connections; only `tt:` storage labels
Layout and devices	Screen fill on six sizes; nothing cut off; laptop card (including touch-screen laptops); sideways card; zoom only on text pages
Steady screen	Back gesture, no long-press menus, no dragging
Visits and feedback	Counting only on the official address and only after the welcome note; Global Privacy Control; switch confirmations; feedback sends only the message, the version and a one-time anti-spam token; size and daily limits
Take A Tour	Every scene on the right screen; controls, drag and speed; nothing tapped or saved; install steps per phone; small screens

Security review (28 Oct 2026)

- Attack test: every saved item was filled with booby-trapped text designed to inject code, then all 29 screens were opened. Nothing ran, nothing broke, no errors.
- No outside connections from any screen, apart from deliberate ones: the anonymous counting script, the feedback box, and Cloudflare's Turnstile anti-spam check on the feedback box.
- Personal-information scans of every file and the full history before publishing.

Defects found and fixed

Defect	Found by	Fix and prevention
Hidden Comets: tiles on the easiest level didn't open with a tap	My phone testing	Fixed; real finger-tap test added
Music silent on the very first tap on phones	My phone testing	Listens for every kind of first touch; test added
Labels shifted on wide phones	Screenshot review	New screen-fill rules; test on six sizes
"Turn upright" card looked out of place	My design review	Redesigned in the garden's style
App wouldn't reopen after the first visit on the live host	My use of the test site	Offline copy stored under the page's real address; local server now mimics the host; test added
Laptop card missing on touch-screen laptops	My own laptop	Detection changed to "main pointer is a mouse or trackpad"; test added
Settings stayed unlocked after leaving it	My review	Locks on leaving; only Settings taps keep it open; test added
Second same-day release labelled "update 3"	Automated test	Label fixed
Page titles hidden behind the panel on short laptop screens	My laptop screenshot	The panel now moves down with the title; test on three sizes
The feedback Send button looked broken during its pause	My preview testing	A 10-second pause with a message explaining it
An older What's New line was cut off at 200 characters	My update on the live site	Accepted (only seen when updating from an old version); the build now refuses any new line over 200 characters.
Thin blurred band at the top and bottom on tall iPhones	My phone testing	Zoom to fill and trim only empty side margins, with the picture's own edge colours filling any remaining band; tested on more screen sizes

Acceptance

- A release is accepted only when all automated tests pass, I have reviewed the screens, I have approved the changes and the "What's New" text, and I have checked it on the protected preview. Only then is it merged and published.

If a release goes wrong

- Roll back: Cloudflare → the Pages project → Deployments → the last good live version → ⋯ → “Rollback to this deployment”. It’s live within a minute.
- People get the restored version through the usual ↑ Update button; nothing they’ve saved is touched. Phones that were offline get it the next time they’re online.
- Fix properly: the fix goes through the normal test branch, preview and my approval.

Known gaps in testing

- Automated tests run in Chrome-type browsers (Chrome, Edge, Android). iPhone Safari was checked by hand on a real iPhone: install, games, Loved Ones, backup and restore, offline use and the tour.
- Game boards haven’t been fully tested with screen readers (VoiceOver, TalkBack).
- No formal colour-contrast check; some text sits on painted pictures and may be harder to read for some people.
- English only.

8. Privacy, Security and Risk Assessment

Privacy by design

- No account, no cookies, no personal tracking, and no server for personal data. There is no central store of personal data to steal. Visits are counted anonymously (Cloudflare Web Analytics) and can be switched off; feedback is anonymous.
- Application data vs hosting data: everything the app saves stays on the phone. Like any website, the hosting service handles basic technical information (such as IP address and browser type) to deliver the site; this is stated plainly on the Privacy & Safety page.
- The Privacy & Safety page was self-reviewed by the creator and Claude against what the app actually does. This is not a legal review.

What is saved, where, and how it is protected

Data	Where	Protection
Contacts (name, number, colour, optional photo)	Browser storage on the phone	Numbers only shown in Settings, behind the passcode
Passcode	Never stored	Only a salted fingerprint
Game progress, stats, stars, music choice	Browser storage on the phone	Not sensitive
Backups	A file the person saves wherever they choose	Encrypted with their password
Visit-counting choice; welcome note seen	Browser storage on the phone	Not sensitive; included in backups
Feedback messages	Cloudflare D1 (hosting service)	Anonymous: time, app version and message only; read from time to time; abusive messages may be deleted

Threats and mitigations

Threat	Mitigation
Malicious code in the page	Strict security rules; all text shown as plain text; attack test passed
Hostile backup files	Every file treated as untrusted and re-checked
Someone else using the phone	Settings passcode; auto-lock on leaving; growing wait after wrong tries
Leaving the app by accident	Back gesture stays inside; tips for Guided Access / App pinning
Look-alike copies	One official address named on the Privacy page

Threat	Mitigation
Personal data in the public code	Scans of every file and the full history before publishing
Copying the work	Visible maker's mark, code notices, invisible artwork watermark, and dated proof of every release
Feedback spam	Accepted only from the app's own address; a hidden honeypot field and Cloudflare Turnstile block automated spam; 10,000-character limit; 300 messages a day in total
Someone takes over my GitHub or hosting account	Stronger sign-in (two-factor or passkeys) on GitHub and Cloudflare; main branch protected once the repository is public
Someone changes the live site without approval	A daily automatic check compares the live site with the approved release and emails me if anything differs

Residual risks (accepted and disclosed)

Risk	Why it remains	Likelihood	Impact
Numbers are readable in browser storage by a technical person with access to the unlocked device	Loved Ones must open WhatsApp with one tap, without the passcode; encryption would mean a passcode for every call; anyone with the unlocked phone could already see these numbers in its own Contacts or WhatsApp	Low	Medium
No recovery without a backup	By design, there is no copy anywhere else; reminders and clear advice mitigate it	Medium	Medium
The phone’s own Home gesture can’t be blocked	No web app can stop the phone’s own swipe home. Guided Access (iPhone) and App pinning (Android) lock the screen to one app; the Privacy & Safety page explains how. Nothing is lost.	High	Low
One-time picture download (about 5 MB, up to 7 MB on older phones)	Needed for offline play; explained on the Privacy page	High	Low

Risk	Why it remains	Likelihood	Impact
Inline styles allowed in the security rules	Needed to place buttons on the paintings. At most, someone could change how a page looks, never run code; and all text people type is shown as plain text.	Low	Low
Preview addresses	Protected by a sign-in so only the owner sees unreleased versions	Low	Low
Anonymous counting uses Cloudflare's script	Off on request, never with Global Privacy Control, never on test previews	High	Low
Tapping a contact gives WhatsApp that person's number	Needed to open the chat; stated plainly on the Privacy & Safety page	High	Low
Someone types personal details into feedback	People are clearly asked not to; anything sent stays in my private database unless I delete it.	Low	Medium

Likelihood: Low = needs unusual circumstances; Medium = will happen to some people; High = happens to most people, or every time. Impact: Low = inconvenience only, nothing personal lost or exposed; Medium = personal data lost, or seen by someone with the phone in hand, but recoverable or limited; High = sensitive data exposed widely, or real harm. No risk here is rated High impact.

Feedback content

- People can type anything into a free-text box: kind words, criticism, or occasionally something hurtful or personal. Feedback is never shown inside the app, so nothing one person writes can reach anyone else. I read it from time to time, may delete anything abusive, and may share short quotes without anything that identifies the writer (the Privacy & Safety page says so). Because it's anonymous, I can't block an individual sender. The protections are a request to keep it kind and leave out personal details, messages accepted only from the app, a hidden honeypot field and a Cloudflare Turnstile check that block automated spam, a 10,000-character limit and at most 300 messages a day. These never limit honest feedback: a real person can write as often as they like. The aim is only to stop automated spam from flooding the box and crowding out real people, and free tools can reduce that without stopping a determined attacker completely. I chose not to filter messages automatically: people are asked not to share personal details, and some may want to leave a way to reach me.

9. Product Decision Log

Each entry: the decision, the options, my choice, and the trade-off I accepted.

#	Decision	Options	My choice	Trade-off accepted
1	Make a public version	Keep private · public version	Separate public app, no personal data, no sign-in	Built and maintained separately
2	Name	Several options	Taarala Thota (Garden of Stars)	Unfamiliar to non-Telugu speakers; the English subtitle explains it
3	Sign-in	Accounts · none	No sign-in	No sync between devices
4	Measuring use	None · full analytics · anonymous counts	Anonymous visit counts (Cloudflare Web Analytics), on by default with a one-time notice; switch on Privacy & Safety and in Settings; Global Privacy Control respected	Approximate numbers only
5	Backups	None · password-locked file	Password-locked file (added after the first version)	Lost password = unusable backup
6	Settings access	Hidden press-and-hold · passcode	Passcode, behind the Settings gear on Home	One more thing to remember
7	Loved Ones visibility	Locked · open names, locked numbers	Open names and photos; numbers behind the passcode	Anyone with the phone can call saved people
8	Contact photos	Names only · photos	Optional photos, redrawn by the app	Slightly more storage
9	Erase wording	Generic · exact	List exactly what is and isn't erased	Longer text
10	Hosting	Several	Cloudflare Pages	Tied to Cloudflare
11	Public history	Squash · full history	Full history, swept for personal data first	Every step visible, including early ones
12	Who commits	AI · me	I commit and push myself, one step at a time	Slower; full control
13	Screen fill	Side strips · fill	Fill the screen without	More layout rules

#	Decision	Options	My choice	Trade-off accepted
			cutting painted content	to test
14	Hints	Every level · Easy only	Easy only	Harder levels have no help
15	Bubble Pop speed	Speed options compared side by side	Moderate speed	Less challenge at level 1
16	Steady screen	Allow gestures · block stray touches	Block stray touches	No zoom on game screens
17	Privacy wording	Simple claim · precise	Separate app data from hosting data	Longer, more honest
18	Code documentation	Top-of-file index · inline notes	Inline notes + developer guide	More to keep current
19	Turning sideways	Squeeze layout · card	Calm card, game paused, redesigned to match the garden	Can't play sideways
20	Ownership	One mark · all three	Visible mark, code notices, invisible watermark	More to maintain
21	Credit	Name only · name + username	Name + GitHub username	Personal name is public
22	Maker's mark	Nine designs reviewed	Constellation seal + signature + shooting star	—
23	Home title artwork	Repaint · keep	Keep the painted title	—
24	Positioning	Specific audience · everyone	A calm space for everyone	Narrower story
25	Laptop visitors	Block · ignore · card	Welcome card with QR and "Continue On This Screen"	One extra tap on laptops
26	Touch-screen laptops	Treat as tablets · as laptops	As laptops	—
27	Tablets	Card · none	No card (they look good upright)	Sideways tablets show a narrow view
28	Bubble Pop levels	Unlock in order · all open	All open, gold stars for goals	Less "earn your way" challenge
29	Updates	Automatic · person	Person chooses, with	Some people stay on older versions

#	Decision	Options	My choice	Trade-off accepted
		chooses	What's New	longer
30	Update after "Later"	Reminder only · always available	Settings section + small Update button, only while an update waits	One more element on screen while waiting
31	Settings update box	Bottom · top	Top, compact, above the passcode	Passcode moves down slightly
32	What's New length	Cap at 5 releases · no cap	No cap, smaller text, scrolls	Long lists for people who skip many updates
33	What's New content	Latest only · everything since your version	Only releases newer than your version	—
34	Release approval	Ship when tested · approve wording too	I approve every change and its What's New text first	Slower releases
35	Settings lock	Timer only · lock on leaving	Lock on leaving; only Settings taps keep it open	Passcode needed more often
36	Zoom	Never · text pages only	Text pages only	Games still can't be zoomed
37	Picture download	Silent · disclosed · on demand	Disclosed on the Privacy page, offline kept	Uses mobile data once
38	Shortcomings	Hide · disclose	Disclose in the docs	Visible limitations
39	Address	Own domain · free address	taarala-thota.pages.dev	Tied to Cloudflare
40	Release process	Test site + upload · test branch	Test branch → protected preview → merge	Extra steps per release
41	Preview access	Public · protected	Protected by sign-in	Sign-in when testing
42	Documentation	Many files · one document	One document + README + developer guide	One long document
43	Feedback	Google Form · own box	Own anonymous box inside the app (no shareable link)	A small server piece
44	Counting consent	Ask · notice	Notice, not a question	The note and an easy switch keep it transparent.
45	Guided tour	Step cards · video file · playing	Playing walkthrough on the real screens, with	More build work

#	Decision	Options	My choice	Trade-off accepted
		walkthrough	pause, drag and speed	
46	Tour wording on privacy	Point to opting out · neutral	Neutral: “adjust a few settings to suit you”	—
47	Install guidance	Generic · per phone	Per phone (Chrome or Safari; both if unknown), with a note that menus can differ	Menus may still look different
48	Capitals	Mixed · Title Case	Title Case on every button and title	—
49	Working rule	Build, then show · show, then build	Brainstorm and mockups first; build after approval	Slower start, fewer rebuilds
50	Using feedback	Private only · may quote	May share short anonymous quotes; read from time to time, with no promise to review daily	Must remove anything identifying before quoting
51	Honesty about limits	Legal section · plain note	“About This Project” note on Privacy & Safety, the README and this document	Admits possible mistakes openly
52	Origin story	Detailed · general	“Safe, calm games for my family”, without naming the private app	Less detail about the private project
53	Licence	All rights reserved · free for non-commercial use · open source	All rights reserved, viewing welcome; provided “as is”	Others must ask before reusing anything
54	App icon	Cut from the Home picture · a new icon	A new painted icon (tree, lanterns, moon, lake)	More detail than ideal at the smallest sizes
55	Hidden Comets artwork	Keep the mine and flag · comet and star	Comet and star	—
56	Filtering feedback	Hide numbers, emails and slurs · leave as is	Leave as is; people are asked to be kind and not share personal details	Some rude or personal content may arrive
57	Messaging loved ones	Build messaging · link to WhatsApp	One-tap link to WhatsApp: people already use and trust it, and the app never sends	Depends on Meta’s app; WhatsApp sees the number when tapped

#	Decision	Options	My choice	Trade-off accepted
			anything itself	
58	Tour video	One full video · a short teaser plus the full tour	A 59-second teaser for social media (some scenes cut, 1.25× speed) and the full tour for the README	The teaser leaves out the install steps
59	Privacy notices	A pop-up every time · quieter notices	A short note in Settings, a once-a-day reminder and a Privacy & Safety page	People must open the page to read the details
60	Minesweeper name	Keep "Minesweeper" · rename	"Hidden Comets", with comets instead of mines ("Minesweeper" is a Microsoft trademark)	A less familiar name for the game
61	Protecting saved numbers	Encrypt contacts inside the app · rely on the phone's own lock and encryption	Rely on the phone's lock; numbers shown only behind the Settings passcode	Numbers rely on the phone's own lock and encryption rather than a second layer in the app; anyone who already has the unlocked phone could find them, just as they could in the phone's own Contacts or WhatsApp.
62	Words in pictures	Paint text into the artwork · add all text in code	All words are added by code; pictures have blank signs	Every picture needs blank signs, but wording can change without repainting, which made the Title Case update easy
63	Keeping contacts	Keep them only in memory (wiped on every refresh) · save them on the phone	Save them on the phone, with password-locked backups: a middle ground, so nobody re-types the same details every visit, and the data never reaches me	Contacts live only on that phone; without a backup they can be lost

#	Decision	Options	My choice	Trade-off accepted
64	Protecting the code	Scramble (obfuscate) the code · readable code with ownership marks	Readable code in a public repository, protected by an all-rights-reserved licence, visible and hidden marks, and dated proof of every release. Scrambling only helps if the code is hidden, and the public repository is meant to show how the work was done.	Anyone can read the code; the marks and dated proof show it's mine, but can't stop copying
65	Play limits	A daily play limit and game timers · none	No daily limit or move timers; only a 60-minute window to resume Chess and Checkers	A family space only works if everyone uses it, and limits might put people off; people manage their own time
66	Who it's for	An open-source template each family sets up for itself · one ready-to-use app	One ready-to-use app for everyone. Once the shared group was dropped (row 67), there was nothing left for a family to set up, and setting up a template is too much work for most people.	Families can't customise or host their own copy
67	Playing and talking with each other	A closed group where an admin adds members and everyone can message and play each other · a one-way list of each person's own loved ones	A one-way list: each person keeps their own loved ones on their own phone, with no sign-in. A shared group would mean holding other people's data and taking responsibility for it, it's far more complex to build for a first version, and someone's loved ones aren't always loved ones to each other, so not everyone would want to be in one group. I may	No playing against each other and no group; each person only connects one-to-one with the people they choose

#	Decision	Options	My choice	Trade-off accepted
			explore it again later.	
68	Encouraging good moves	None · praise every strong move · only clear-cut good moves	A brief "Nice move!" that fades on its own, shown only on clear-cut good moves (a capture that wins material, or reaching a king/promotion), in Chess and Checkers at every level | Subtle positional moves are not praised, so it never praises a weak move by mistake; it appears after you move, so it is encouragement, not a hint	
69	Reviewing a finished game	None · keep a history of finished games · replay from the game's saved moves	After a game ends, an optional walkthrough replays the moves on a clean board; it closes on Back or Home and is not stored on its own, and the moves it replays stay on the device with the resumable game and clear when the game ends	The moves stay on the device, never sent anywhere; once a game ends there is no lasting history to revisit
70	Tall phones (e.g. iPhone)	Leave blurred strips · zoom to fill and trim the sides · fill the edge with the picture's own colours	Zoom in to fill the screen, trimming only empty side margins (never a painted sign), and fill any thin remaining band with the picture's own edge colours	On a very tall phone a sliver of the far left and right edges is trimmed
71	Protecting the feedback box from spam	Nothing extra · limit how often each person can write · block automated spam only	Block automated spam only — a hidden honeypot field and a Cloudflare Turnstile check, on top of the existing app-only rule and daily cap — and never limit a real person, who	Free tools reduce automated spam but cannot stop a determined attacker completely; rarely, a real person on a very fast or

#	Decision	Options	My choice	Trade-off accepted
			can write as often as they like	unusual setup may need to tap Send again